Reg. No. [ ] [14EC46]

FOURTH SEMESTER B.TECH UNIVERSITY EXAMINATION, JUNE – 2016

ELECTRO MAGNETIC FIELD THEORY

Time: 3 Hours Maximum Marks:70

Instructions:

1. **Answer any FIVE full questions by choosing at least 2 Questions from each part.**
2. **Any missing data may be suitably assumed.**

PART - A

1. a. Explain Coloumb's law. Draw the typical variation of force between two changed entities and distance between them. **(04+02=06)** 06

   b. Given that $\bar{A}=30e^{-\rho}\bar{a}_\rho-2z\bar{a}_z$ in a cylindrical coordinates. Evaluate both sides of divergence theorem for the volume enclosed by ρ=2, z=0 and z=5 comment, whether this function could be created in a laboratory environment. 08

2. a. Infinite number of charges of 1 nC are placed along x axis at x=1, 2,4,8 ... ∞. Find the electric potential at point x=0, due to all charges. Is this LTI system? (6+2=8) 08

   b. A bakelite box is placed on top of a wooden block. Derive the boundary conditions for electric field between the two. 06

3. a. State and prove uniqueness theorem. 06

   b. Use Laplace's equation to find the capacitance per unit length of a coaxial cable of inner radius 'a' and outer radius b. Assume $v=v_0$ at r=a and v=0 at r=6. Also calculate the ratio (b/a) to achieve a capacitance of 10μF/m with air dielectric. (6+2=08) 08

4. a. State and prove Stoke's theorem. Also compare divergence theorem and stokes theorem. (6+2=08) 08

   b. $\bar{A}=x^2\bar{a}_x+2yz\bar{a}_y-x^2\bar{a}_z$. Find $\bar{B}$ Also, comment on number of dimensions required to visualize this. (5+1=06) 06

PART – B

5. a. A point charge of $Q=-1.2C$ has a velocity $\bar{v}=(5\bar{a}_x+2\bar{a}_y-3\bar{a}_z)ms^{-1}$ Find the magnitude of force exerted on charge, find f. 08

   i. $\bar{E}=-18\bar{a}_x+5\bar{a}_y-10\bar{a}_z\ Vm^{-1}$

   ii. $\bar{B}=-4\bar{a}_x+4\bar{a}_y+3\bar{a}_z\ T$

   b. Derive an expression for inductance of a solenoid. Identify the parameter which rapidly increases the self-inductance. (4+2=06) 06

6. a. Write Maxwell's equations in differential form. Explain how did James Clerk Maxwell repair Ampere's circuital law for time varying fields. 04

b. Given $E = E_m \sin(\omega t - \beta z)\bar{a}_y$. Sketch E and H with appropriate reasoning. 04

Also mention the total number of dimensions to visualize a typical linearly polarized electromagnetic wave vector field. 02

7. a. Obtain solution of wave equation for a uniform plane wave in free space. 10

b. Malaysian Airlines sunk deep into the ocean floor. Explain why electromagnetic engineers failed to detect this gigantic metallic aircraft with a typical microwave radar. 04

8. a. A uniform plane wave of 200MHz travelling in free space impinges normally on a large block of dielectric, FR4 epoxy having $\varepsilon_r=4$ $\mu_r=9$ and $\sigma = 0$ Calculate transmission and reflection coefficients at the interface. 10

b. Define 04

i. Reflection coefficient

ii. Standing wave ratio